

AGYEI HAGGLA MENSAH

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RESEARCH INTERESTS

Cybersecurity and Secure Systems: Privacy-preserving technologies, mobile and digital-financial security, applied cryptography, secure software design, authentication, data protection, and security for resource-constrained environments. Interested in investigating how real-world systems expose security and privacy risks through metadata, authentication workflows, data persistence, and system architecture.

Broader interests: Distributed systems, cloud data protection, human-computer interaction, and software engineering.

EDUCATION

University of Energy and Natural Resources (UENR), Sunyani, Ghana — Bachelor of Science in Information Technology | Expected Aug. 2026

Relevant Coursework: Data Structures & Algorithms, Database Management, Computer Networks, Software Engineering, System Analysis & Design, Web Development, Data Mining, Computer Crimes & Forensics, Information Security, Human-Computer Interaction, Enterprise Resource Planning

Tuobodom Senior High/Technical School, Ghana — West African Senior School Certificate Examination (WASSCE) | 2020

RESEARCH & TECHNICAL PROJECTS

VoxUnite — University Digital Voting Platform

Full-Stack Developer | MERN + TypeScript

- Built a full-stack university voting platform with separate administrator, student, and real-time monitoring dashboards.
- Implemented OTP-based voter authentication, role-based access control, layered duplicate-vote prevention, audit logging, and real-time election updates.
- Investigated security and privacy trade-offs in digital voting, including voter eligibility, vote integrity, auditability, and ballot-linkability limitations.
- Designed the system using React/TypeScript, Node.js/Express, MongoDB/Mongoose, and Socket.IO.

Charity Foundation Payment System

Full-Stack Developer | MERN + Paystack

- Developed a web-based payment system integrating Paystack for online donation processing.
- Implemented HMAC-SHA512 webhook verification to authenticate payment notifications and protect against forged webhook requests.
- Designed backend services for transaction processing, verification, and payment-state management.

Internship Management System

Lead Developer / Research Topic

- Designed a multi-user internship management system supporting account creation, verification, logging, tracking, and grading.
- Developed administrative functionality for faculty to monitor and assess student internship milestones.
- Applied database and access-control concepts to a multi-role institutional information system.

GradLink — Student-Employer Job Platform

Lead Developer | MERN + Socket.IO

- Designed and deployed a multi-role career platform supporting administrators, applicants, and verified employers.
- Developed role-based dashboards and backend workflows for employer verification and application tracking.
- Implemented real-time communication functionality using Socket.IO.

Additional Projects: Buywise (e-commerce price comparator), Inkwell (MERN blogging platform), Mobile Money Transaction System (C++), CUG Exam Eligibility System.

TECHNICAL EXPERIENCE

IT Intern — Catholic University of Ghana | Fiapre, Ghana | Oct. 2025 – Dec. 2025

- Supported university IT operations across multiple departments, including hardware/software diagnostics, infrastructure configuration, user access management, and security-profile administration.
- Assisted with deployment and maintenance of IT infrastructure and troubleshooting of technical issues.
- Documented technical procedures to improve internal IT knowledge and operational reference materials.

LEADERSHIP & OUTREACH

President — Reaching Out to Every Heart Foundation | Feb. 2024 – Present

Lead and coordinate community-development outreach programs and manage volunteer activities; coordinate planning and execution of community initiatives.

Technology Awareness Facilitator — Seikwa SDA JHS | June 2024

Introduced junior high school students to basic computer skills, internet safety, and data privacy.

Volunteer — Exam Support Team, Catholic University of Ghana | May 2026

Assisted with examination invigilation and classroom allocation.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Elements of AI** — University of Helsinki

- **Introduction to Cybersecurity** — Cisco Networking Academy, Dec. 2025
- **Computer Hardware Basics** — Cisco Networking Academy, Nov. 2025
- **Active Member, IT/Tech Society** — University of Energy and Natural Resources, Feb. 2023 – Present

TECHNICAL SKILLS

Programming: JavaScript, TypeScript, Python, C++, Java (basic), HTML5/CSS3

Frameworks & Systems: React, Vite, Node.js, Express, REST APIs, Socket.IO

Databases: MongoDB, MySQL

Security & Networking: Authentication & authorization, role-based access control, OTP workflows, hashing, HMAC, TCP/IP, LAN/WAN configuration, secure API design

Tools: Git/GitHub, Figma, Adobe Photoshop, Canva, Microsoft Office

Languages: English (Professional), Twi (Native), Kolango (Fluent), French (Beginner)